Day 01 | Thursday

01

October 29, 2026

Whole numbers and estimation

9:00 AM–10:15 AM • Math • 75 min

Calculator basics and checking

10:30 AM–12:00 PM • Calculation Practice • 90 min

Week 1 • 2 scheduled blocks • 165 teaching minutes

Estimate. Show the method. Label units. Check the result.

Today’s exact schedule

9:00 AM–10:15 AM • 75 min

Whole numbers and estimation

9:00 AM–9:05 AM	**Start**	5
9:05 AM–9:10 AM	**Recall**	5
9:10 AM–9:25 AM	**Teach**	15
9:25 AM–9:40 AM	**Together**	15
9:40 AM–10:00 AM	**Practise**	20
10:00 AM–10:10 AM	**Check**	10
10:10 AM–10:15 AM	**Wrap up**	5

Math • Math Week 1, Lesson 1

10:30 AM–12:00 PM • 90 min

Calculator basics and checking

10:30 AM–10:35 AM	**Start**	5
10:35 AM–10:40 AM	**Recall**	5
10:40 AM–10:55 AM	**Teach**	15
10:55 AM–11:10 AM	**Together**	15
11:10 AM–11:45 AM	**Practise**	35
11:45 AM–11:55 AM	**Check**	10
11:55 AM–12:00 PM	**Wrap up**	5

Calculation Practice • Math Week 1, Lesson 2

Break: 10:15 AM–10:30 AM • 15 minutes • excluded from teaching time

Whole numbers and estimation

Today’s goal: Read whole-number quantities; estimate and calculate a cut-list total; explain why an answer is reasonable.

Name one use for today’s skill.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Exercise Book 1A Q1–10 (p.6), Q11–17 (pp.6–7)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Recall and reconnect

Use Exercise Book 1A Q3,5,9 without a calculator. Discuss which operation finds an offcut.

Try first without displayed answers.

Resources: Exercise Book 1A Q1–10 (p.6), Q11–17 (pp.6–7)

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Recall from the Exercise Book | 1A Q3

What is the 7 worth in 37,215?

Ten thousands	Thousands	Hundreds	Tens	Ones
3	7	2	1	5

Write the operation, working and units before checking.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Read the value before calculating

1. A digit’s place tells its value: in 4,072, the 4 means four thousand.
2. Read the quantity and unit together; 1,200 mm is a length.
3. When rounding, name the target place before changing any digits.
4. An estimate uses simpler nearby values; it predicts the size of the answer.

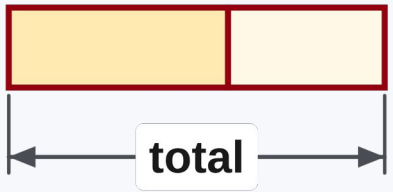
Thousands	Hundreds	Tens	Ones
4	**0**	**7**	**2**
4,000	0	70	2

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

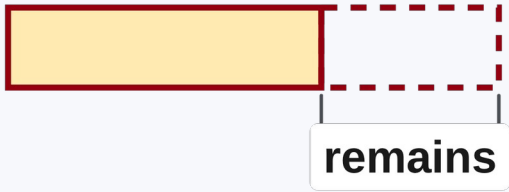
Choose an operation that matches the quantity

1. Add separate quantities to find a total.
2. Subtract the amount used from the starting amount to find what remains.
3. Multiply equal groups; divide to share equally or count groups.
4. Compare the exact result with the estimate and explain any large difference.

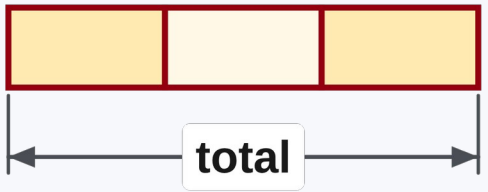
Add: put together



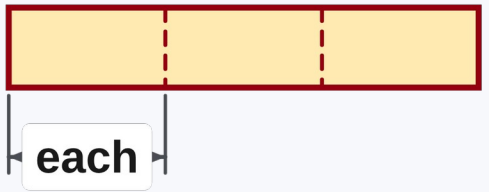
Subtract: what remains



Multiply: equal groups



Divide: share equally



Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

An estimate should agree with the scale

Classroom example: 48 packs contain 21 fasteners each.

Estimate: 50 × 20 = 1,000 fasteners
Calculate: 48 × 21 = 1,008 fasteners

The exact answer is close to 1,000—not 100 or 10,000.
A calculator result still needs a quantity, unit and reason.



Work together

Exercise Book 1A Q11, 13 and 15.
Estimate first; alternate solver and checker. Compare hand and calculator methods.

W1 L1 • Selected calculator: TI-30Xa or Sharp EL-531XT

Textbook | page 14

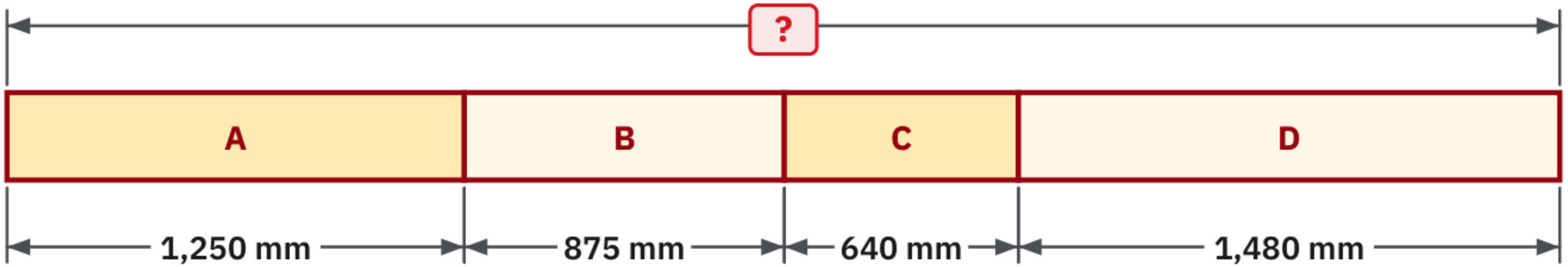


Figure 1.2 The cut list from Example 1.4. The total length is hidden.

Textbook p.14 · Figure 1.2: read the cut-list dimensions.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Textbook | page 14

Example 1.4 Estimating a cut list

A cut list has four pieces: 1,250 mm, 875 mm, 640 mm, 1,480 mm. Estimate the total length.

SOLUTION

1. Round each length to the nearest hundred: 1,300, 900, 600, 1,500.
2. Add the rounded numbers: 1,300 + 900 + 600 + 1,500 = 4,300.

Answer **About 4,300 mm. The exact total, found in Example 1.5, is 4,245 mm, which is close to the estimate.**

Textbook p.14 · Example 1.4: estimate before calculating.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Spot the mistake. Explain the correction.

A learner estimates 48 × 21 as 1,000, then accepts a calculator result of 100. What check failed?

Discuss the method before checking the answer.

The result is about ten times too small. Re-enter the intended multiplication and compare again.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Work together

Work Exercise Book 1A Q11,13,15 together. Estimate first, then compare hand and calculator methods.

Work in pairs; alternate solver and checker.

Resources: Exercise Book 1A Q1–10 (p.6), Q11–17 (pp.6–7)

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Guided practice | 1A Q11

3,275 + 1,948

$$\begin{array}{r} 3{,}275 \\ +\ 1{,}948 \\ \hline \end{array}$$

Work with a partner. Take turns explaining the next step.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Compare your method | 1A Q11

3,275 + 1,948

CHECK TOGETHER

5,223

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Your scheduled practice

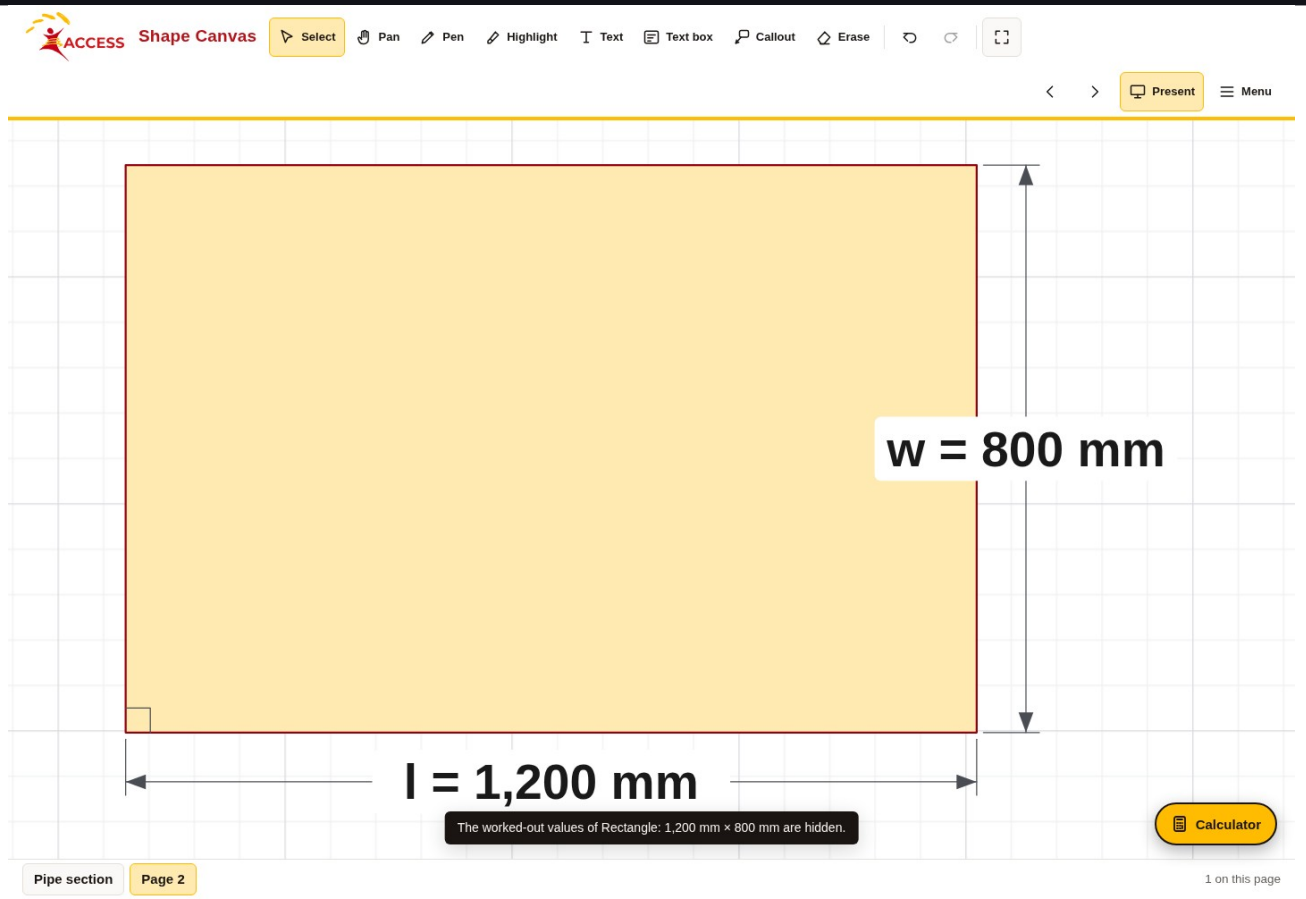
Complete Exercise Book 1A Q1–10 and Q12,14,16,17 as time allows. On Shape Canvas, set a rectangle to 1,200 mm by 800 mm and read both sizes only, since area and perimeter come later in the course.

Show working and units; estimate, then check.

Resources: Exercise Book 1A Q1–10 (p.6), Q11–17 (pp.6–7)

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Open Shape Canvas



Navigation example. Use the values and units in the assigned task.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Practise in Numeracy Hub and Shape Canvas

1. Supported: In Numeracy Hub Calculator, select TI-30Xa or Sharp EL-531XT and clear the display.

2. Independent practice: Estimate and write your assigned 1A calculation before checking its result.

3. Supported preview: In Shape Canvas, label a 1,200 mm × 800 mm rectangle; read both lengths only.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Practise from the Exercise Book | 1A Q1

Write 4,072 in words.

Thousands	Hundreds	Tens	Ones
4	0	7	2

Write the operation, working and units before checking.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Practise from the Exercise Book | 1A Q2

Write in digits: fifteen thousand three hundred eight.

Ten thousands	Thousands	Hundreds	Tens	Ones

Write the operation, working and units before checking.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Check independently

Exercise Book 1A Q5 and Q10; explain one estimate and show the operation used.

Answer independently and show the method.

Resources: Exercise Book 1A Q1–10 (p.6), Q11–17 (pp.6–7)

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Independent check | 1A Q5

Round 3,462 to the nearest ten.

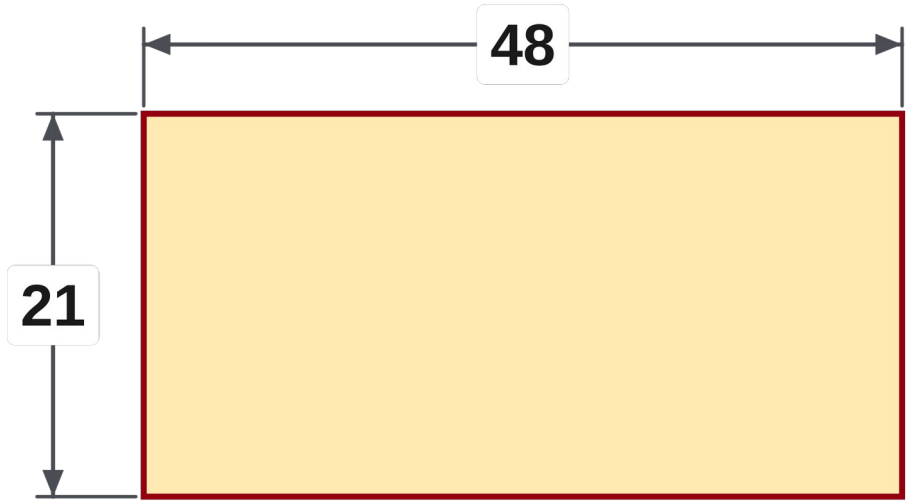
Thousands	Hundreds	Tens	Ones
3	4	6	2

Answer independently. Keep worked solutions closed.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Independent check | 1A Q10

Estimate 48 × 21 by rounding each number to the nearest ten.



Answer independently. Keep worked solutions closed.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Wrap up and choose the next step

Share one learning point and one skill to practise next.

Record a takeaway and next action.

Resources: Exercise Book 1A Q1–10 (p.6), Q11–17 (pp.6–7)

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Break

10:15 AM–10:30 AM

Return ready for: Calculator basics and checking

This break is excluded from teaching minutes.

Calculator basics and checking

Today’s goal: Use the chosen calculator correctly and check an answer with an estimate or inverse operation.

Name one use for today’s skill.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Exercise Book 2A Q1–3,6 (p.17); 1A Q11–17,20 (pp.6–7)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Recall and reconnect

Try to clear the display and enter 23 × 23. Check which model they use.

Try first without displayed answers.

Resources: Exercise Book 2A Q1–3,6 (p.17); 1A Q11–17,20 (pp.6–7)

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

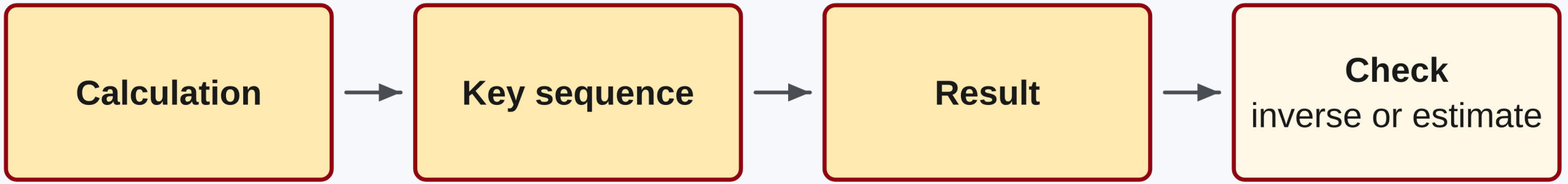
Know what the key will do

1. Clear the old calculation before beginning a new problem.
2. Use the correction key to repair an entry; reread the intended calculation.
3. The square key multiplies a number by itself; square root reverses this for positive values.
4. Change sign reverses a value’s sign; subtraction finds the difference between two values.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Keep a record that can be checked

1. Write calculation → key sequence → result, using your selected calculator model.
2. For TI-30Xa and Sharp, follow the textbook's own square-root entry sequence.
3. Use an inverse operation: multiply a quotient back or square a square-root result.
4. Use an estimate as a separate check that the answer has the expected size.

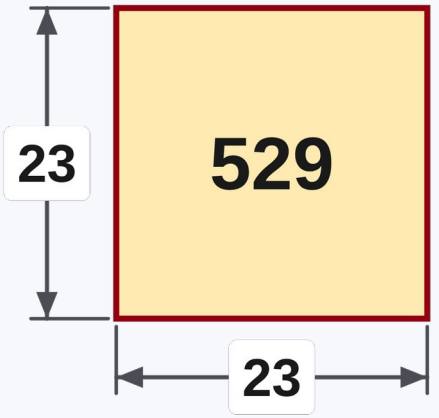


Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Reverse the operation to check

$23^2 = 23 \times 23 = 529$
$\sqrt{529} = 23$

Squaring and square root undo one another for these positive values.
Write the intended calculation before entering keys.



Square
$23^2 = 529$

Square root
$\sqrt{529} = 23$

A second classroom example

2,688 ÷ 12 = 224 pieces
Check: 224 × 12 = 2,688. An estimate near 225 supports the result.

Work together

Exercise Book 2A Q1–3, p.17. Record calculation, keys and answer; compare root entry on the two models.

Textbook | page 49

Job	TI-30Xa	Sharp EL-531XT
Turn on	ON/C	ON/C
Turn off	OFF	2ndF then ON/C (OFF is printed above it)
Second functions	2nd	2ndF
Change the sign of a number	+⇄− after the digits	+/− before the number
Square root	√x after the number	√ before the number, then =
Square	x^2 after the number	x^2 after the number, then =
Store in memory	STO then 1, 2 or 3	STO then a letter key (A to F, X, Y, M)
Fix a mistyped digit	←	DEL

Textbook p.49 · Table 2.1: key differences between the two calculators.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Textbook | page 52

CALCULATOR

TI-30Xa

12^2

12 | x²

Display 144.

√144

144 | √x

Display 12.

Sharp EL-531XT

12^2

12 | x² | =

Display 144.

√144

√ | 144 | =

Display 12.

Answer $12^2 = 144$ and $\sqrt{144} = 12$. Each answer checks the other.

Textbook p.52 · Square and square-root key sequences; each checks the other.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Spot the mistake. Explain the correction.

Does pressing the subtraction key necessarily enter a negative number correctly?

Discuss the method before checking the answer.

No. Use the selected model’s change-sign key for a negative value; use subtraction between values.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Work together

Solve Exercise Book 2A questions 1–3 together. Compare square-root entry on the two models; estimate before checking.

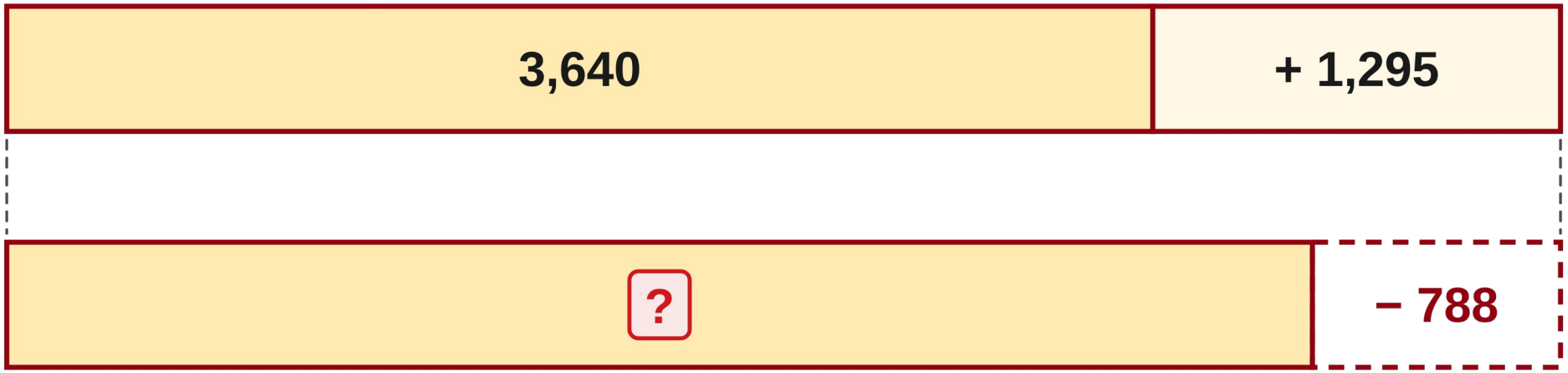
Work in pairs; alternate solver and checker.

Resources: Exercise Book 2A Q1–3,6 (p.17); 1A Q11–17,20 (pp.6–7)

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Guided practice | 2A Q1

3,640 + 1,295 − 788



Work with a partner. Take turns explaining the next step.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Compare your method | 2A Q1

3,640 + 1,295 − 788

CHECK TOGETHER

4,147. TI-30Xa: 3640, +, 1295, −, 788, =. Sharp: 3640, +, 1295, −, 788, =.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Your scheduled practice

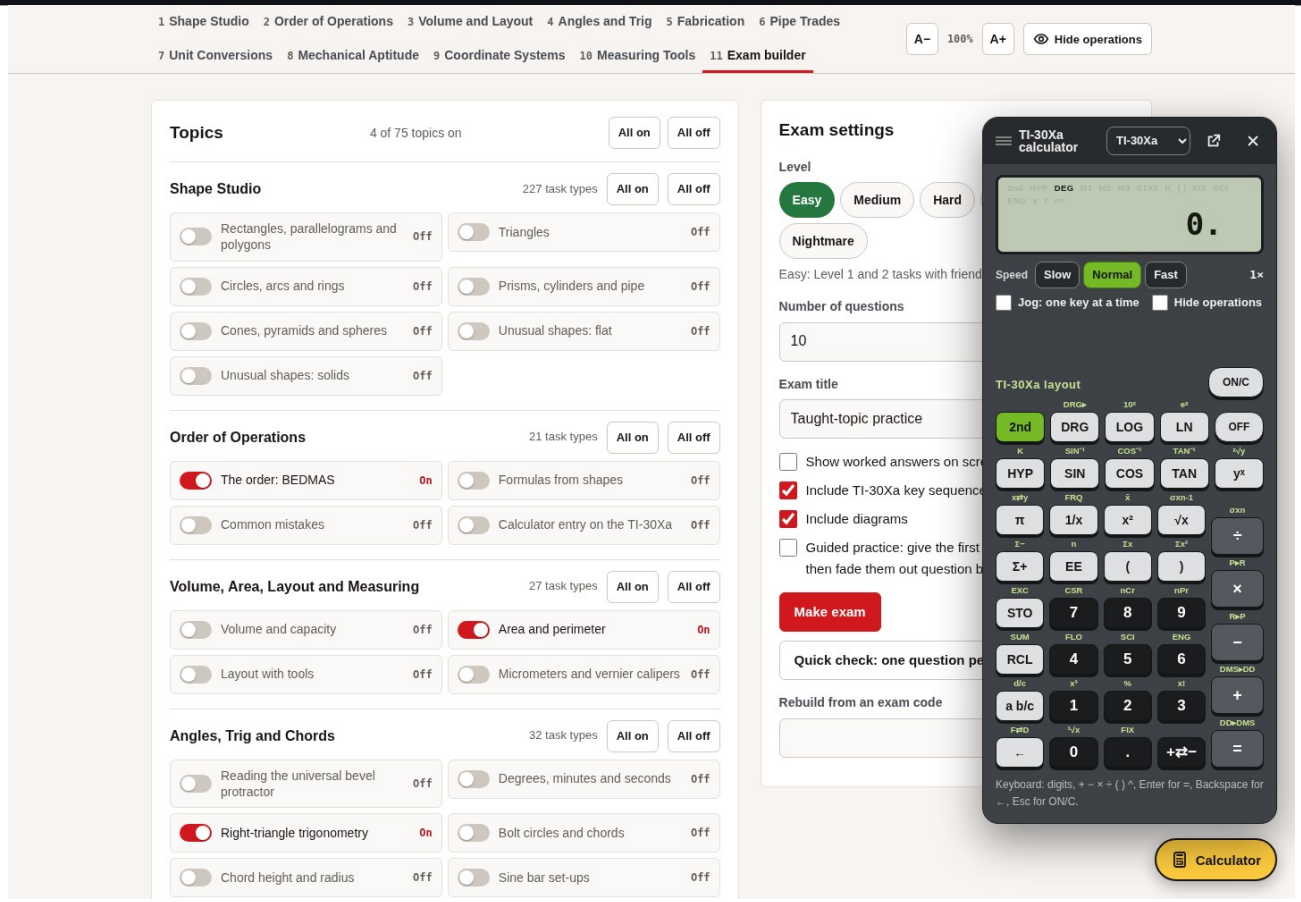
Complete 2A question 6 and selected 1A questions 11–17,20. Record the calculation, keys and answer; use inverse operations to verify two results.

Show working and units; estimate, then check.

Resources: Exercise Book 2A Q1–3,6 (p.17); 1A Q11–17,20 (pp.6–7)

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Open Numeracy Hub



Navigation example. Use the values and units in the assigned task.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Practise in Numeracy Hub

1. Supported: Choose one calculator model in Numeracy Hub; locate clear, square, root and sign keys.

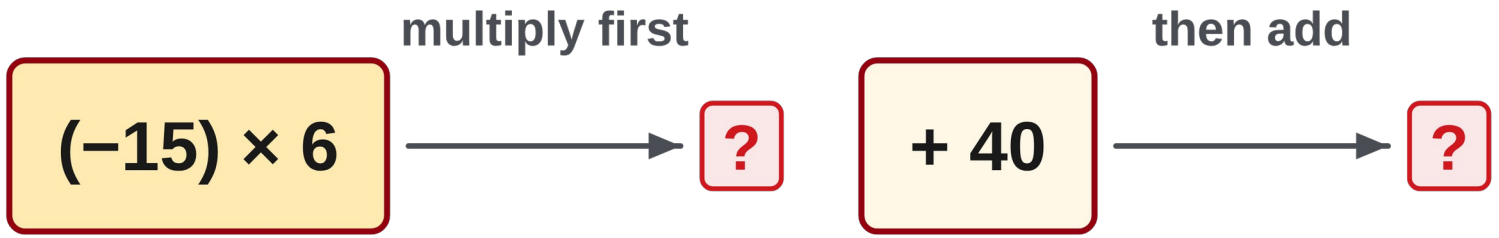
2. Supported: Work 2A Q1–3 and record each model-specific key sequence beside the written calculation.

3. Independent practice: Check two completed calculations using inverse operations; retain the working.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Practise from the Exercise Book | 2A Q6

$(-15) \times 6 + 40$



Write the operation, working and units before checking.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Practise from the Exercise Book | 1A Q11

3,275 + 1,948

$$\begin{array}{r} 3{,}275 \\ +\ 1{,}948 \\ \hline \end{array}$$

Write the operation, working and units before checking.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Check independently

Redo Exercise Book 2A questions 1 and 3 independently. Explain one checking method.

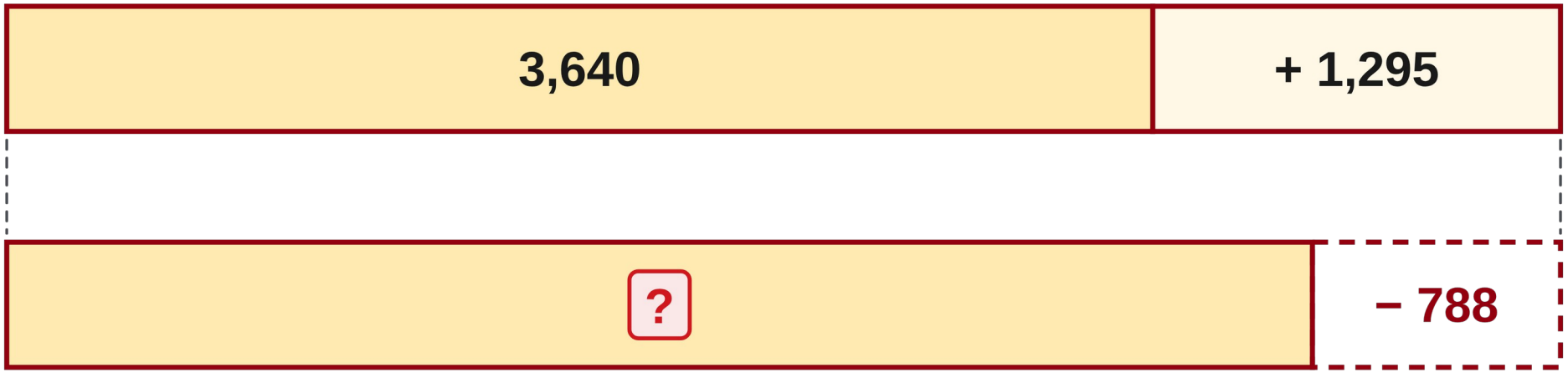
Answer independently and show the method.

Resources: Exercise Book 2A Q1–3,6 (p.17); 1A Q11–17,20 (pp.6–7)

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Independent check | 2A Q1

3,640 + 1,295 − 788

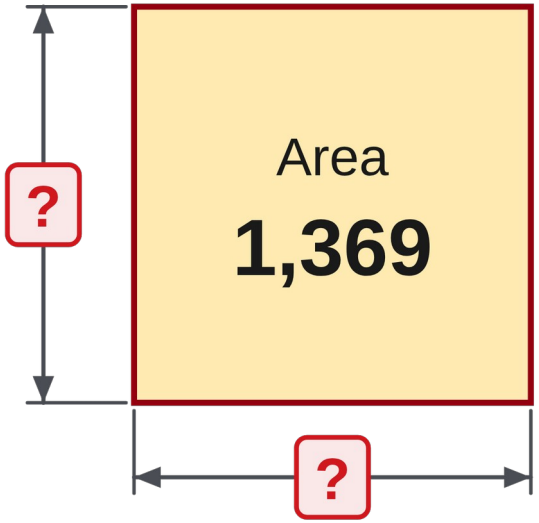


Answer independently. Keep worked solutions closed.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Independent check | 2A Q3

√1,369



Answer independently. Keep worked solutions closed.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Wrap up and choose the next step

Share one learning point and one skill to practise next.

Record a takeaway and next action.

Resources: Exercise Book 2A Q1–3,6 (p.17); 1A Q11–17,20 (pp.6–7)

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23